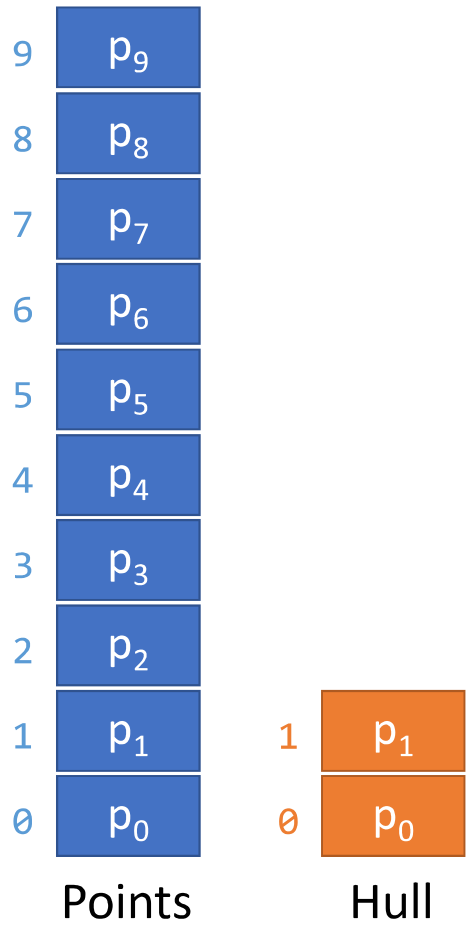
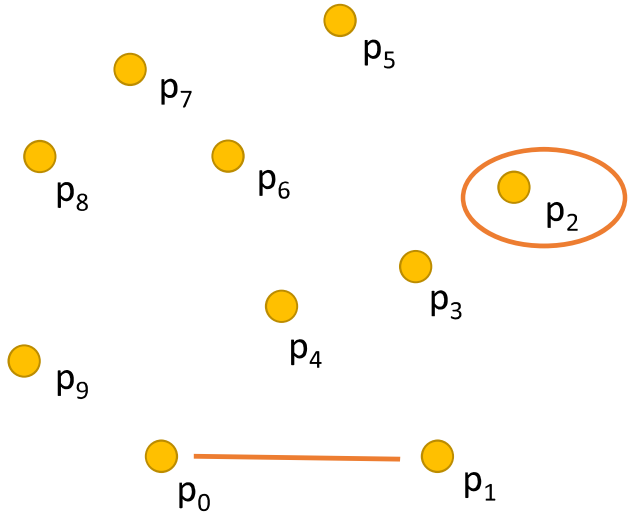
The Graham scan step by step

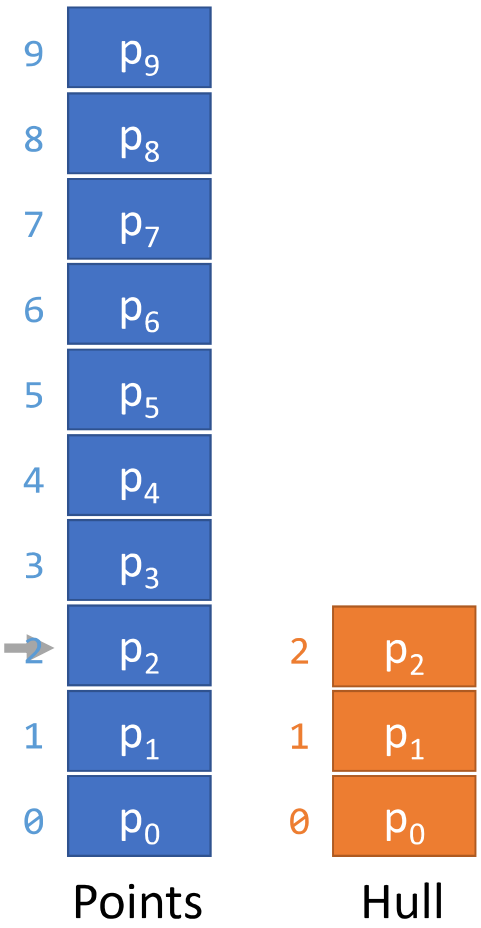
```
Initalise:                    add first two points to Hull list        # on the hull

                              n = 11        # length of points list
                              a_pos = 1     # start position for a in triplet apq when i=2
```





```
for... i = 2
```

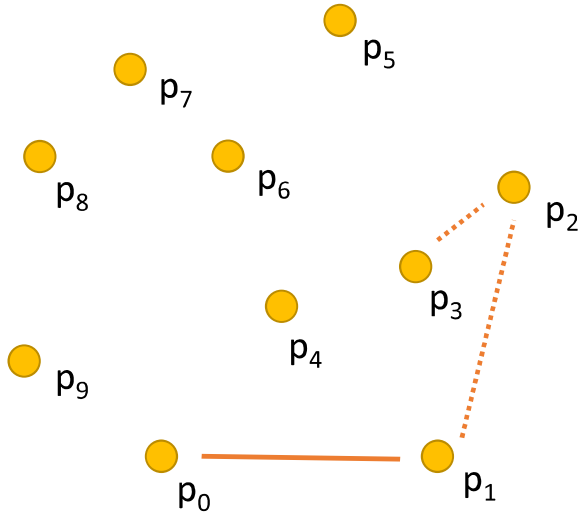



```
i = 2
a_pos = 1
```

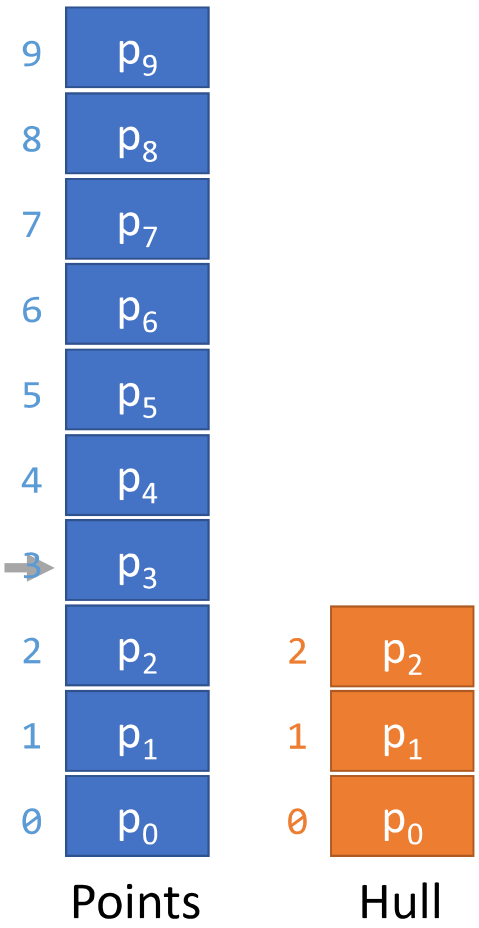
---

a = p_1, **p = p_2**, q = p_3

```
apq = left turn so p is on the hull
    hull.append(p2)
    a_pos = i
```



```
for... i = 3
```

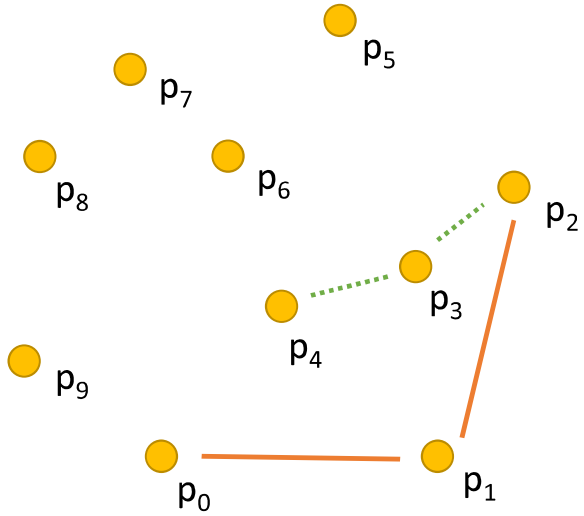



```
i = 3
a_pos = 2
```

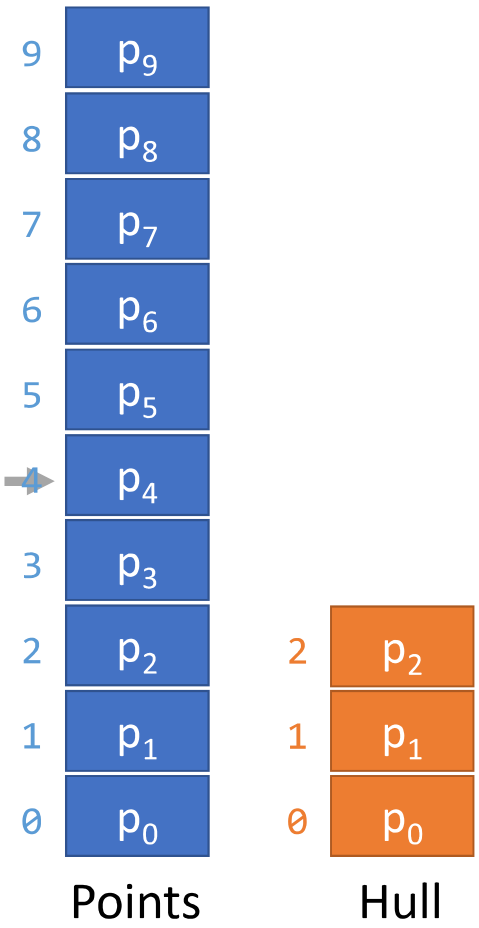
---

a = p_2, **p = p_3**, q = p_4

```
apq = right turn so p is not in polygon
hull remains unchanged
a_pos remains unchanged, p and q advance
```



for... i = 4

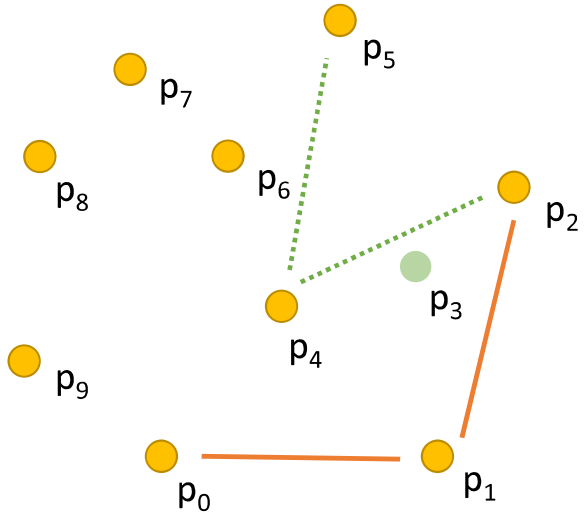


i = 4
a_pos = 2

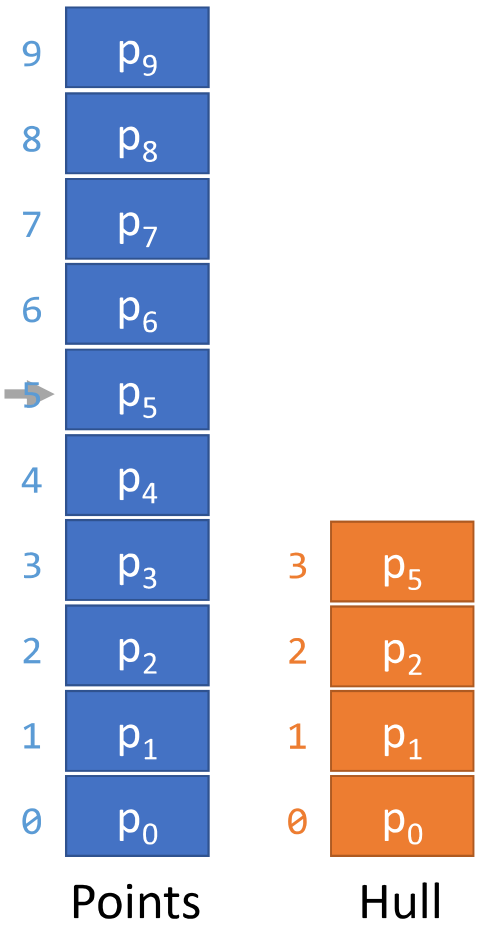
---

a = p_2, **p = p_4**, q = p_5

apq = right turn so p is not in polygon
hull remains unchanged
a_pos remains unchanged, p and q advance



```
for... i = 5
```

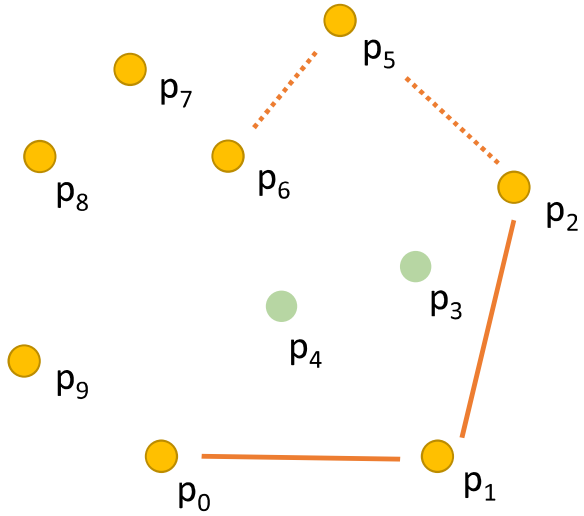



```
i = 5
a_pos = 2
```

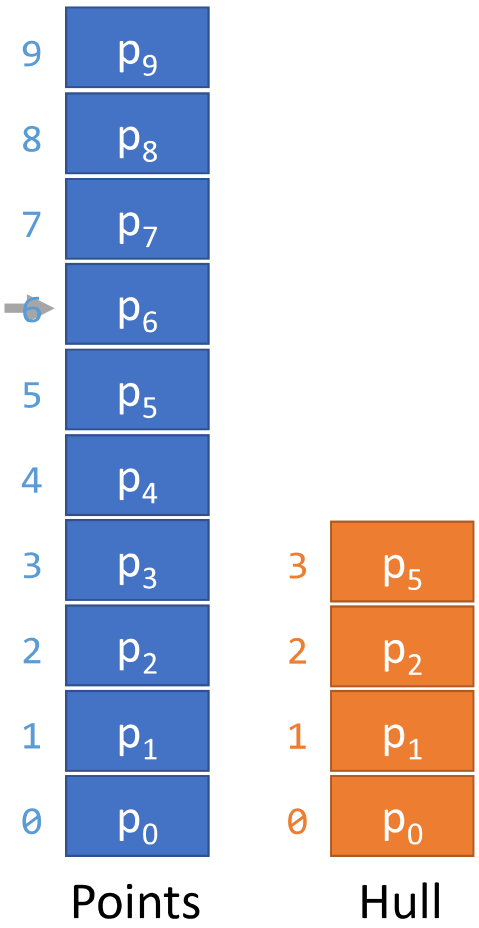
---

a = p_2, **p = p_5**, q = p_6

```
apq = left turn so p is on the hull
    hull.append(p5)
    a_pos = i
```



```
for... i = 6
```

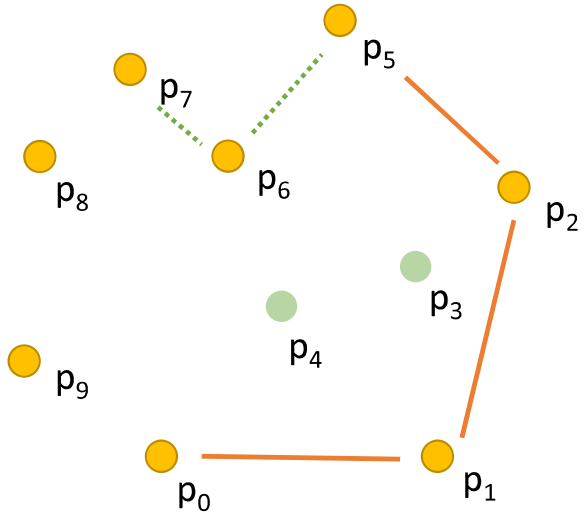



```
i = 6
a_pos = 5
```

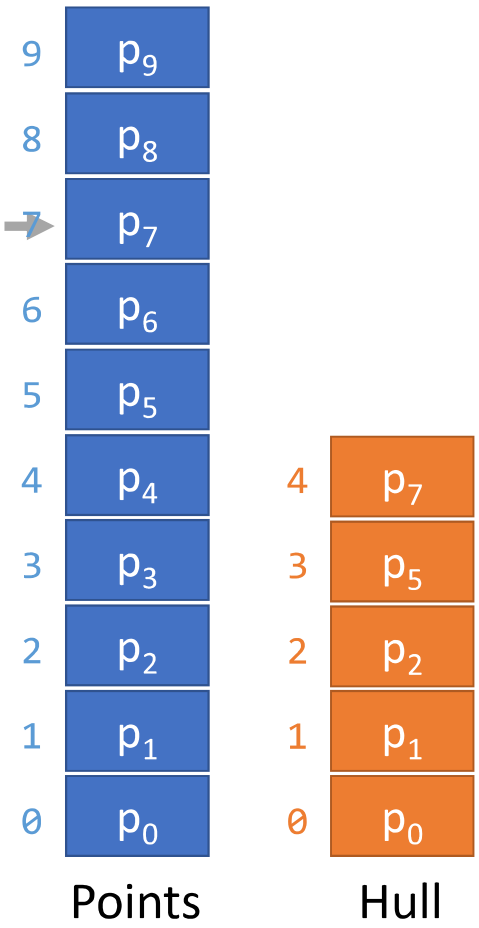
---

a = p_5, **p = p_6**, q = p_7

```
apq = right turn so p is not in polygon
hull remains unchanged
a_pos remains unchanged, p and q advance
```



```
for... i = 7
```

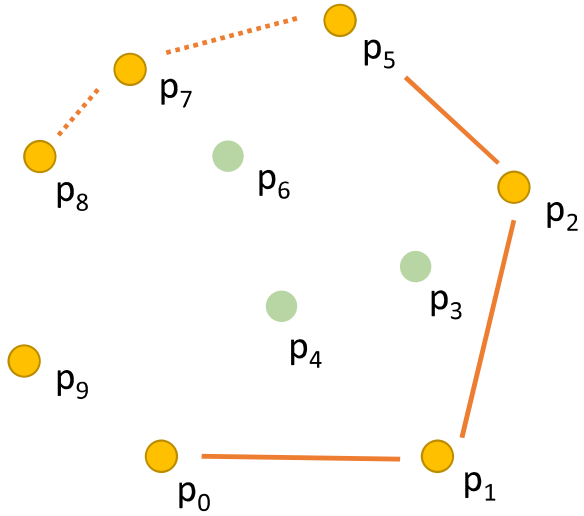



```
i = 7
a_pos = 5
```

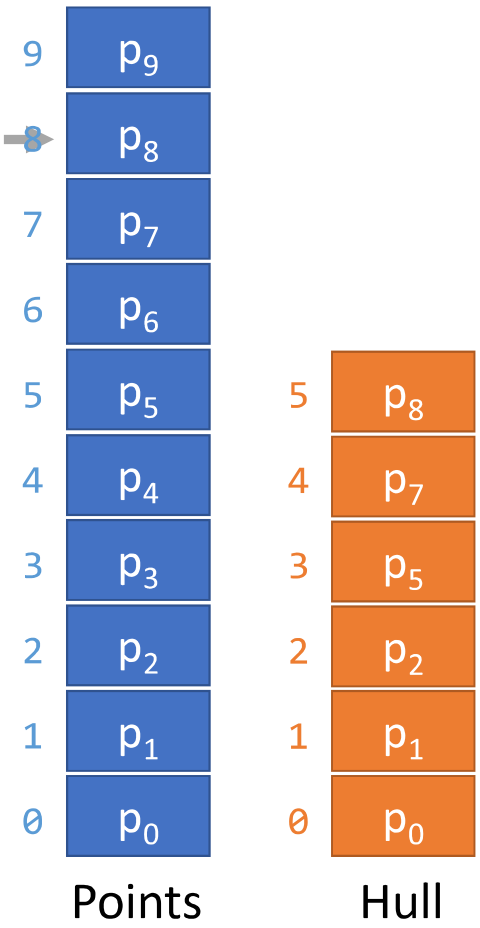
---

a = p_5, **p = p_7**, q = p_8

```
apq = left turn so p is on the hull
    hull.append(p7)
    a_pos = 7
```



```
for... i = 8
```

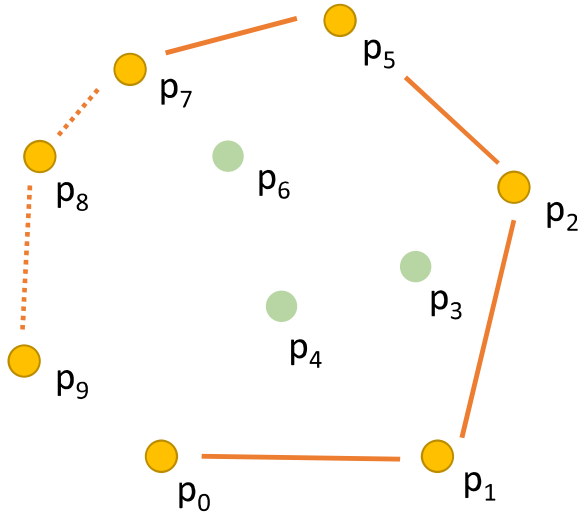



```
i = 8
a_pos = 7
```

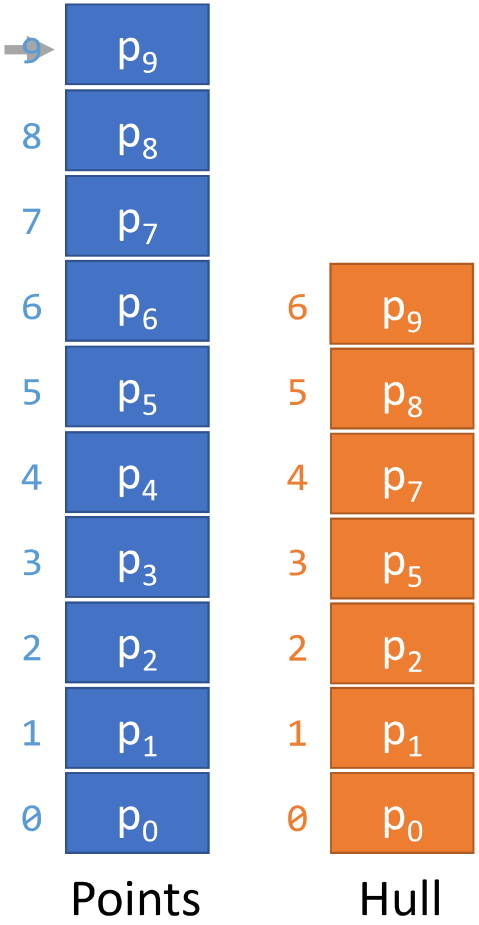
---

a = p_7, **p = p_8**, q = p_9

```
apq = left turn so p is on the hull
    hull.append(p8)
    a_pos = 8
```

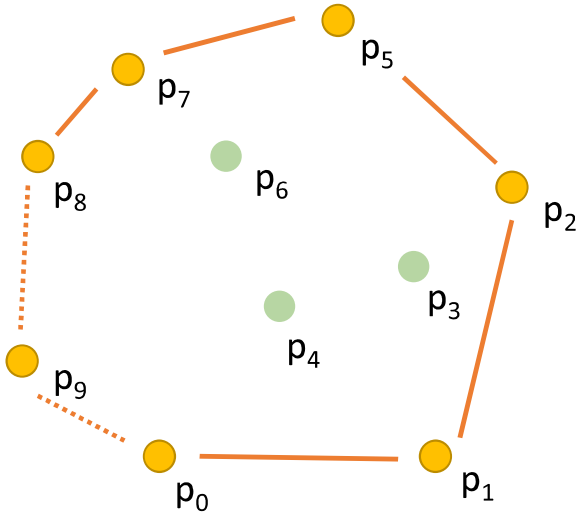


```
for... i = 9
```

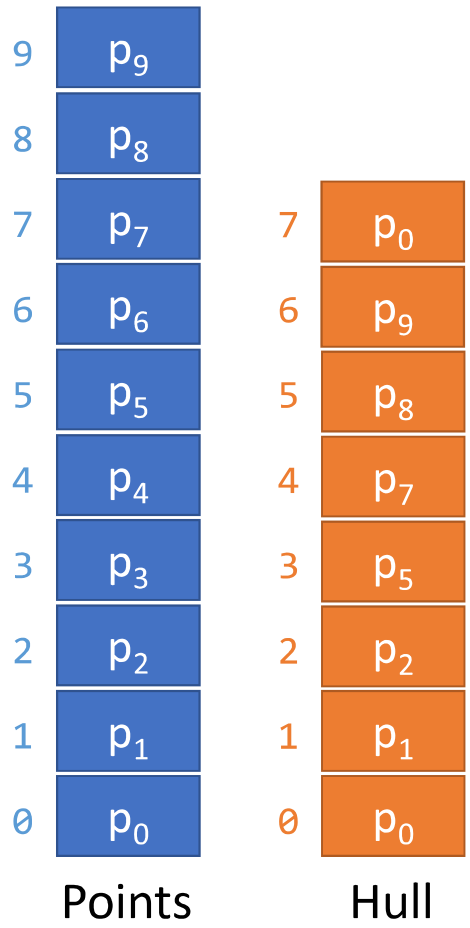


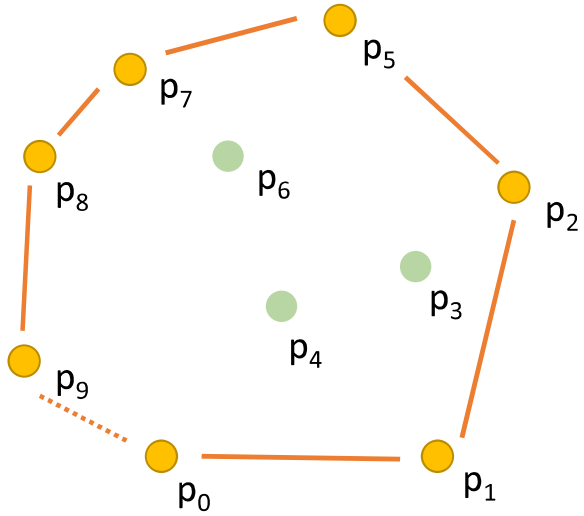

```
i = 9
End of loop
```

```
hull.append(p9)  # last point will be on the hull
```



Finish: `hull.append(p₀)  # copy start point to close the polygon`

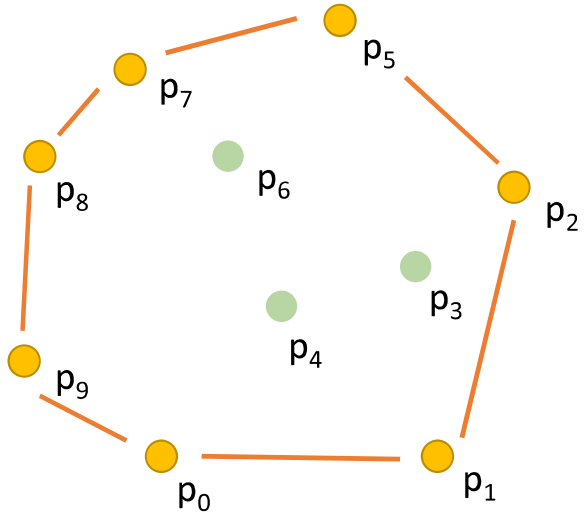




Implementing the Graham Scan method

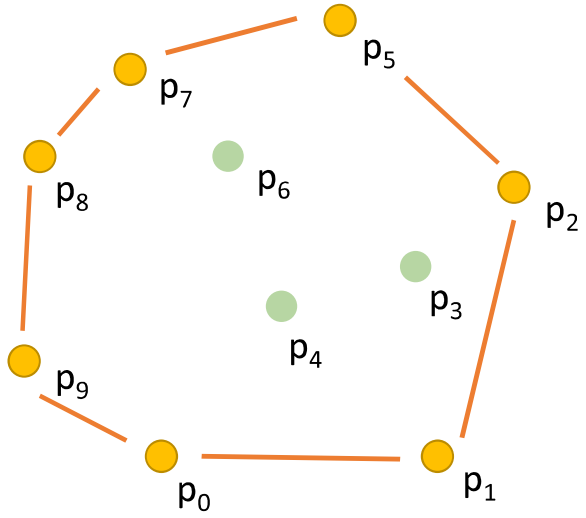
	Points
9	p_9
8	p_8
7	p_7
6	p_6
5	p_5
4	p_4
3	p_3
2	p_2
1	p_1
0	p_0

	Hull
7	p_0
6	p_9
5	p_8
4	p_7
3	p_5
2	p_2
1	p_1
0	p_0



Graham Scan efficiency

- This part of the algorithm is (obviously) ~ O(n)
- What about the sorting? If use QuickSort or similar, sorting has an efficiency of O(nlogn)
- So, the overall efficiency of Graham Scan is defined by the sort

	Points
9	p_9
8	p_8
7	p_7
6	p_6
5	p_5
4	p_4
3	p_3
2	p_2
1	p_1
0	p_0

	Hull
7	p_0
6	p_9
5	p_8
4	p_7
3	p_5
2	p_2
1	p_1
0	p_0